4. Group Project (Group 3)

Reflection

EGR 3350-07

Technical Communications for Engineers and Computer Scientists

Ms. Alysoun Taylor-Hall

Utsav Acharya | Computer Science | April 30, 2026

Reflection: Improving the Design of Floating Solar Farms

Working on the Group 3 proposal to improve the efficiency and durability of floating solar farms in ocean environments was the most collaborative writing experience of the semester. The group consisted of five members: Carter Rockenbaugh, Daniel Evans, Erica Gilliam, Reese Harmeyer, and me, where Erica served as the team lead, helping coordinate the division of tasks and keeping the project on track.

From the beginning, our team divided the work according to each member's background and strengths. Carter focused on site selection and cost feasibility. Daniel handled the materials research, specifically the selection of high-density polyethylene (HDPE) and the anti-corrosive epoxy glass coating. Erica researched the mooring and anchoring system. My contribution centered on the software-based IoT monitoring and detection system, which aligned with my Computer Science background. I researched how sensors could continuously track panel performance and trigger the automated cleaning response when efficiency dropped below a set threshold. Reese works on cleaning robots.

The group worked well together overall. Meetings were productive, and we always left with clear next steps. The proposal came together better than I initially expected. The sections integrated smoothly because we agreed on a shared outline and consistent formatting before each person began writing independently. The LOI received a score of 7.7 out of 8.0 (A), and the instructor's feedback praised the work while asking us to clarify our exact scope further.

The most challenging part for me personally was bridging the technical depth of the monitoring system research with the need to keep the proposal accessible to a mixed engineering audience. I had to describe IoT sensor systems, voltage and current tracking, and automated alert mechanisms in a way that Carter and Erica could follow without a software background. This required me to think carefully about audience adaptation, one of the core skills emphasized throughout the course.

Overall, this project demonstrated that group writing succeeds when members communicate consistently, respect each other's expertise, and agree on structure early. I came away with a stronger understanding of how collaborative technical documents are built and a greater appreciation for the role that clear division of labor plays in producing a cohesive final product.